



June 17, 2026

Dear PhD Admissions Committee,

I am excited to apply for the "Resource-Efficient Quantum Simulation of Chemistry with Quantum Computers" PhD programme at the University of Southampton, located in Southampton, United Kingdom.

As a data science engineer and AI enthusiast, I am drawn to this project because it combines my passion for machine learning and quantum computing. My research interests align perfectly with the goals of this project, as evident from my previous work on "Medical *MambaTS* : *linear – time statespace model for sub – millisecond bedside inference*". *This achievement has direct implications for resource – efficient quantum algorithms, as the MambaTS algorithm can be adapted to efficiently simulate molecular behavior.*

Furthermore, my experience with "PI *DeepONet* : *0D Windkessel Physics – Informed Operator Network using PyTorch Autograd*" *has given me a deep understanding of the importance of efficient and physically meaningful.*

I am confident that my background in physical sciences and programming, combined with my prior experience in quantum computing, make me an ideal candidate for this project. I am excited about the opportunity to work on developing resource-efficient quantum algorithms for molecular simulations and their implementation in quantum hardware.

Thank you for considering my application. I look forward to discussing my qualifications further.

Sincerely,

Ismail Aissaoui